Calculate gravitational force

antocciphysics

UNIT 2 / DAY 5 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate a gravitational interaction with consistent units.

SUCCESS CRITERIA - I can...

1. Identify both masses and the center separation.
2. Square the distance before dividing.
3. Check units and the scale of the result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A ball begins with downward velocity 3 m/s and falls 2 s. Use downward positive, g=10 m/s² and no drag. Find final velocity and displacement.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F = Gm_1m_2/r^2$; $G = 6.67\times10^{-11}$ N·m²/kg².
r is center-to-center separation. CPA may use the supplied numerator.

Worked reasoning: In a supplied gravity model, $Gm_1m_2=18$ N·m² and r=3 m. Square first: $r^2=9$ m², so $F=18/9=2$ N, attractive.

Your turn: Sketch the two centers, label r and add the two force arrows. Show how the units reduce to newtons.



Marzano check - circle: ① ② ③

03 Read and interpret the model

For spheres 4 m apart center to center, a student substitutes r=2 m because each is 2 m from the midpoint. Identify the error.

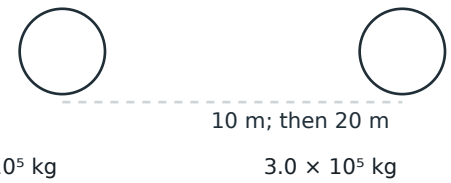


Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

Two idealized spherical space objects have masses 2.0×10^5 kg and 3.0×10^5 kg. Their centers are 10 m apart. Use $G = 6.67 \times 10^{-11}$ N·m²/kg². Label the model and calculate their attraction. Repeat at 20 m.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why can a tiny force between classroom-sized objects be physically real even if you cannot feel it?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

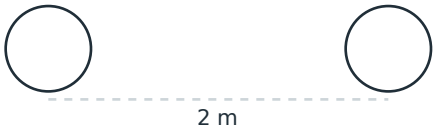
A 4 kg box slides left at 6 m/s on a rough horizontal floor after release. Draw forces and explain what follows about vertical and horizontal acceleration.
Draw the forces on the chosen object and label their sources.

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

Two 1000 kg spheres have centers 2 m apart. Use $G = 6.67 \times 10^{-11}$ N·m²/kg². Calculate the attraction. A classmate divides by r instead of r²: by what factor does that change the numerical answer?



1000 kg 1000 kg

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A pair has force 0.12 N at 2 m. Without finding either mass, calculate Gm_1m_2 and the separation for a force of 0.03 N.

Separation (m)	Force (N)
___	___
___	___

Marzano check - circle: 1 2 3